

Bayesian Inference

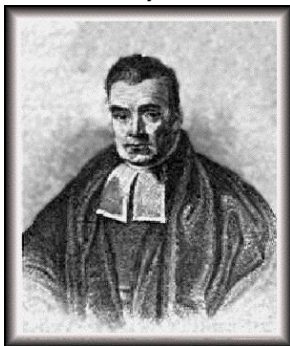
STAT 4530/6530
Statistical Inference for Data Scientists

Introduction to Bayesian Inference

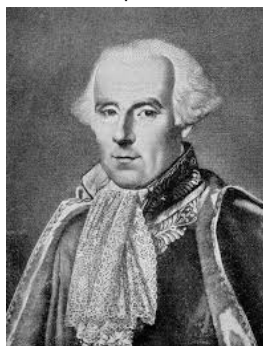
What is Bayesian statistics?

- An alternative line of statistical reasoning inspired by [Thomas Bayes](#), who formulated what is known as **Bayes' Theorem** in the mid-1700s
- [Pierre-Simon Laplace](#) popularized the field of Bayesian statistics from the late 1700s to early 1800s

Bayes



Laplace



Probability

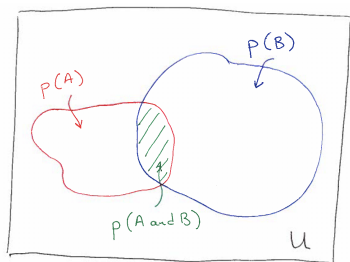
Consider a *sample space* U , which is the set of all possible outcomes, and *events* A and B , which are subsets of the sample space

Example: We want to record whether a person can speak French and/or Dutch.

- Let F and F' respectively denote whether the person can or cannot speak French
- Let D and D' respectively denote whether the person can or cannot speak Dutch
- Let event A be that the person can speak French, and let event B be that the person can speak Dutch
- The sample space is then $U = \{(F, D), (F, D'), (F', D), (F', D')\}$
- The events are $A = \{(F, D), (F, D')\}$ and $B = \{(F, D), (F', D)\}$
- We can also consider the event that the person speaks French and Dutch:
 $A \cap B = \{(F, D)\}$

Conditional Probability

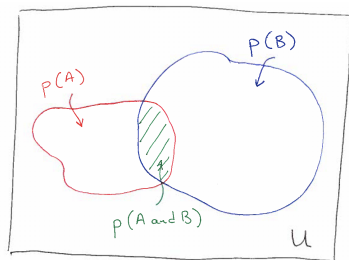
Given a sample space U , and events A and B , consider the marginal probabilities $P(A)$ and $P(B)$, and the joint probability $P(A, B)$ (the probability of A **and** B)



Example: Suppose we have a sample of 100 residents of Brussels: 69 speak French but not Dutch, 4 speak Dutch but not French, and 27 speak French and Dutch

- $P(A) = (69 + 27)/100 = 0.96$
- $P(B) = (4 + 27)/100 = 0.31$
- $P(A, B) = 27/100 = 0.27$

Conditional Probability



We can express $P(A, B)$ in terms of marginal probabilities $P(A)$ and $P(B)$ and conditional probabilities $P(A|B)$ (the probability of A **given** B) and $P(B|A)$ (the probability of B **given** A):

$$P(A, B) = P(A)P(B|A) = P(B)P(A|B).$$

Substituting the probabilities from our example in the identity above gives us:

$$\frac{27}{100} = \frac{96}{100} \frac{27}{96} = \frac{31}{100} \frac{27}{31}.$$

Bayes' Theorem

Starting with the expression of $P(A, B)$ in terms of marginal and conditional probabilities,

$$P(A, B) = P(A)P(B|A) = P(B)P(A|B)$$

we can rearrange the terms and solve for the conditional probability $P(A|B)$, which gives us Bayes' Theorem:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}.$$

In terms of random variables X and Y (functions defined on U), we can express Bayes' Theorem as:

$$P(X|Y) = \frac{P(Y|X)P(X)}{P(Y)}.$$

Bayesian Inference

Consider observed data Y and model parameters θ related by the sampling distribution $P(Y|\theta)$.

In Bayesian inference, we want to infer the distribution of the unknown parameters θ from the observed data Y .

Bayes' Theorem makes this possible:

$$P(\theta|Y) = \frac{P(\theta)P(Y|\theta)}{P(Y)}$$

Two ingredients required to specify a Bayesian statistical model:

- **Sampling Distribution/Likelihood:** $P(Y|\theta)$ – describes belief that we observe data Y if the true parameter value is θ (also required in “classical” statistical inference)
- **Prior distribution:** $P(\theta)$ – describes prior belief about the distribution of θ

Bayesian Inference

Holy grail in Bayesian inference: **posterior distribution** $P(\theta|Y)$, where

$$P(\theta|Y) = \frac{P(\theta)P(Y|\theta)}{P(Y)}$$

- Describes our **updated** beliefs about θ after observing data Y
- We can answer most inferential questions once we obtain $P(\theta|Y)$

How do we specify $P(\theta)$?

- Choose a distribution that describes the prior information or beliefs that we have about θ
- In case we do not have much in the way of prior information or beliefs about θ , we can use a vague or diffuse prior that characterizes the lack of prior information

Example

Suppose we wish to estimate the proportion θ of Brussels residents whose native language is French

Data: from a sample of $n = 100$ Brussels residents, $Y = 79$ report a native language of French

Sampling Distribution: If Y is a random variable counting the number of “successes” (Brussels residents reporting French as their native language) in a sample of size n , and θ represents the true rate of “success” in the Brussels population, an appropriate sampling model is

$$Y|\theta \sim \text{Binomial}(n, \theta),$$

where the binomial distribution probability mass function is:

$$P(y|\theta) = \binom{n}{y} \theta^y (1 - \theta)^{n-y} \text{ for } y = 1, \dots, n$$

Example: Classical Inference

After specifying the sampling model, we can proceed with classical inference

Maximum Likelihood Estimate: $\hat{\theta} = 79/100 = 0.79$

A single point estimate $\hat{\theta}$ does not account for the uncertainty in estimating the true value θ

Estimate to quantify uncertainty: 95% confidence interval (CI) for θ

Under our model, this interval is:

$$\left(\hat{\theta} - 1.96 \sqrt{\frac{\hat{\theta}(1 - \hat{\theta})}{n}}, \hat{\theta} + 1.96 \sqrt{\frac{\hat{\theta}(1 - \hat{\theta})}{n}} \right)$$

With $n = 100$ and $y = 79$, we get 95% CI (0.71, 0.87)

Interpretation of Confidence Intervals

What is the meaning of a 95% confidence interval?

In this example, it means that there is a 95% probability that θ is between 0.71 and 0.87. Right?

Interpretation of Confidence Intervals

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WRONG! The 95% confidence interval does **NOT** mean that there is a 95% probability that θ is between 0.71 and 0.87

Interpretation of Confidence Intervals

Classical inference framework:

- θ is an unknown parameter, but its value is fixed
- The confidence interval either contains θ or it does not
- **Meaning of 95% CI:** suppose we randomly sample 100 Brussels residents, record how many are native French speakers, and construct a 95% CI as described. If we repeat this procedure a large number of times (with different 95% CIs constructed from different samples), the percentage of the 95% CIs that contain the true value of θ will be close to 95.

Bayesian perspective:

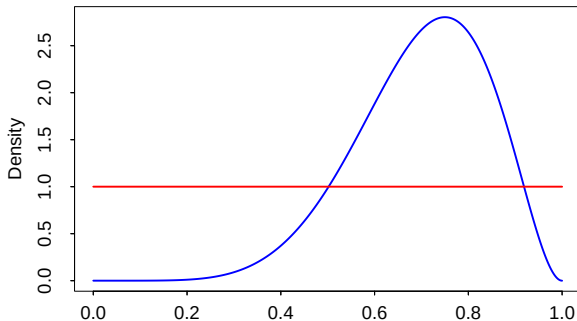
- Parameters are unknown random variables that have distributions
- A statement such as “there is a 95% probability that θ is between 0.71 and 0.87” would make sense in a Bayesian framework.

Bayesian Inference: Simple Example

Model specification for Bayesian inference requires the extra step of specifying a prior distribution for the model parameter θ

Prior 1: We do not have any specific prior information or beliefs about θ , and we can express this lack of prior information by assigning θ a Uniform[0, 1] distribution, which posits that any value between 0 and 1 is equally plausible

Prior 2: We can adopt a more informative Beta(a, b) prior with mean $\frac{a}{a+b}$ and variance $\frac{ab}{(a+b)^2(a+b+1)}$. For example, we can specify $a = 7, b = 3$:

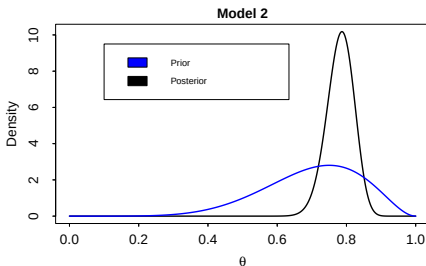
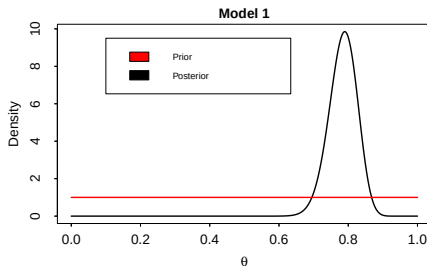


Bayesian Inference: Simple Example

Model 1: Under Prior 1, the posterior distribution is $\theta|y \sim \text{Beta}(80, 22)$. From the quantiles of this distribution, we can compute a 95% Bayesian credible interval (BCI) of (0.70, 0.86).

- **Interpretation of 95% BCI:** conditional on the observed data y , there is a 95% probability that θ is between 0.70 and 0.86.

Model 2: Under Prior 2, the posterior distribution is $\theta|y \sim \text{Beta}(86, 24)$ and the 95% BCI is (0.70, 0.85).

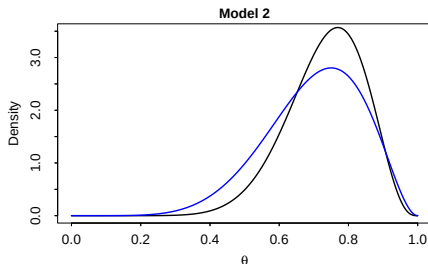
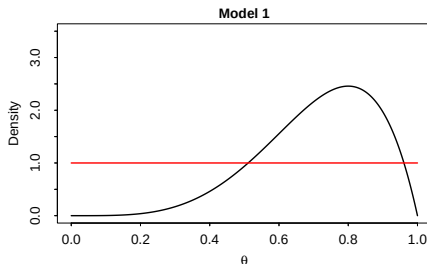


Example with Less Data

Suppose that we instead of a smaller sample of size $n = 5$ with $y = 4$ native French speakers. Because the data provide less information, the prior has more influence on the posterior:

Model 1: Under Prior 1, the posterior distribution is $\theta|y \sim \text{Beta}(5, 2)$, and we get a 95% BCI of (0.36, 0.96).

Model 2: Under Prior 2, the posterior distribution is $\theta|y \sim \text{Beta}(11, 4)$ and the 95% BCI is (0.49, 0.92). The more informative prior yields a posterior that is less spread out.



Bayesian Inference

Goal in Bayesian modeling is to obtain the posterior distribution

- The posterior distribution reflects the combination of the prior and the observed data
- The amount of influence the prior and data have on the posterior depends on how informative they are relative to each other.
- There is not necessarily a “right” or “wrong” prior distribution. Different prior distributions reflect different prior beliefs and lead to different Bayesian models.
- It is common practice to perform sensitivity analyses with a range of different priors.

Bayesian Inference for the Binomial Model

The Basics

Prior Distributions

Recall that in a Bayesian framework, the prior distribution $p(\theta)$ expresses the prior knowledge or beliefs about the model parameter(s) θ .

In practice, the level of information provided by prior distributions varies. Prior distributions span the range between being highly informative' to "noninformative."

- Highly informative – the prior provides information that is crucial to the estimation of the model
- "Noninformative" – the prior expresses as little information about θ as possible so as to allow the likelihood to dominate in the model estimation

Weakly Informative Priors

It is often of interest to specify a prior distribution for a parameter that provides little or no information about the values assumed by that parameter.

- This is often accomplished via a prior distribution that spreads its mass approximately evenly across a wide range of values rather than having its mass concentrated in a more narrow range.
- Such priors are known as *vague* or *diffuse* or *weakly informative* priors

Female happiness example

[from Hoff, Section 3.1] A sample of 129 females aged 65 or older were asked whether or not they were generally happy in the 1998 General Social Survey. Of these, 118 reported being generally happy, while the remaining 11 reported not being generally happy.

Notation:

Let n denote the sample size, and y the number of females who reported being generally happy

We have $n = 129$ and $y = 118$

Goal: Estimate the true rate θ of generally happy females aged 65 or older.

Ingredients for Bayesian modeling

Every Bayesian analysis requires specification of the following:

- **Sampling model:** $p(y|\theta)$

How likely is it that we observed y females generally happy in our sample of size n , given that the true happiness rate is θ ?

- **Prior model:** $p(\theta)$

Summarize beliefs associated with different values of the true happiness rate θ , prior to observing the data.

The posterior distribution

Our goal: study the **posterior distribution**, denoted $p(\theta|y)$

- Interpretation: $p(\theta|y)$ summarizes beliefs associated with different values of true happiness rate θ AFTER observing data
- Given the prior $p(\theta)$ and sampling model $p(y|\theta)$, we can compute the posterior by **Bayes' Theorem**:

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)}$$

Sampling model

Let Y be a random variable counting the number of “successes” within a sample of size n . If θ represents the **true rate of “success”** in the population, then an appropriate sampling model is:

$$Y|\theta \sim \text{Bin}(n, \theta)$$

Probability mass function (pmf):

$$p(y|\theta) = \binom{n}{y} \theta^y (1 - \theta)^{n-y} \text{ for } y = 1, \dots, n$$

Note: This step of specifying an appropriate sampling model would also be undertaken in a classical statistical inference framework

“Noninformative” prior model 1

Suppose the only prior belief we have about θ is that any value between 0 and 1 is *equally plausible*. An appropriate, “noninformative” prior model is:

$$\theta \sim \text{Unif}[0, 1]$$

Probability density function (pdf):

$$p(\theta) = 1 \text{ for } 0 \leq \theta \leq 1$$

Note: Because we are now interpreting probabilities as degrees of belief, it makes sense (and is necessary) to talk about probability distributions for θ , whereas this would have no merit from a frequentist perspective.

Computing posterior 1

Given our sampling and prior model 1, we can use **Bayes' Theorem** to directly compute the **posterior distribution**:

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)},$$

where $p(y) = \int_{-\infty}^{\infty} p(y|\theta)p(\theta) d\theta$ is the *normalizing constant* (also called the *marginal probability* or *marginal likelihood*)

We can show that the posterior is a **beta distribution**:

$$\theta|y \sim \text{Beta}(y + 1, n - y + 1)$$

Make sure you understand how we can show this!

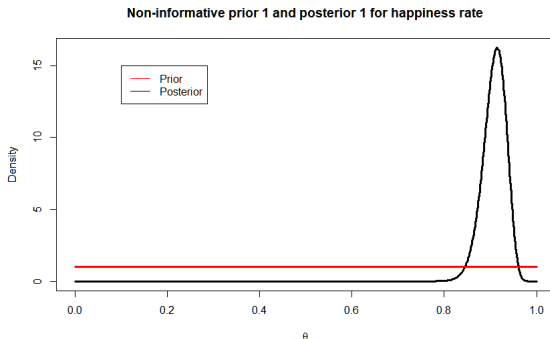
Visualizing posterior 1

Posterior 1 for female happiness example:

$$n = 129, y = 118$$

$$\theta|y = 118 \sim \text{Beta}(y + 1 = 119, n - y + 1 = 12)$$

We can visualize how the observed data updates prior 1 beliefs into posterior 1 beliefs about θ :



Prior model 2

Question: What if we had knowledge of results from a previous General Social Survey, which identified that 45 of 60 females aged 65 or older were generally happy?

An **informative prior** based on this information may satisfy the following:

- **Center:** based on this previous study, we might guess that θ is relatively close to $45/60 = 0.75$. So we may want our prior to be centered here.
- **Spread:**
 - This is a relatively small survey – how “spread out” should our prior for θ be?
 - A larger sample size suggests more “trust” in our prior information and less of a spread in the prior distribution

Prior model 2

If $\theta \sim \text{Beta}(a, b)$ then we can show that $\theta|y \sim \text{Beta}(a + y, b + n - y)$ and

$$\begin{aligned} E[\theta|y] &= \frac{a + y}{a + b + n} \\ &= \frac{a + b}{a + b + n} * \frac{a}{a + b} + \frac{n}{a + b + n} * \frac{y}{n} \\ &= \frac{a + b}{a + b + n} * \text{prior mean} + \frac{n}{a + b + n} * \text{sample mean} \end{aligned}$$

This is a weighted average of the prior mean and the sample mean

Note that n is the sample size of the data and y is the number of “successes” or 1’s observed in the data

So we can interpret

- a as the “prior number of successes”
- b is the “prior number of failures”
- $a + b$ as the “prior sample size”

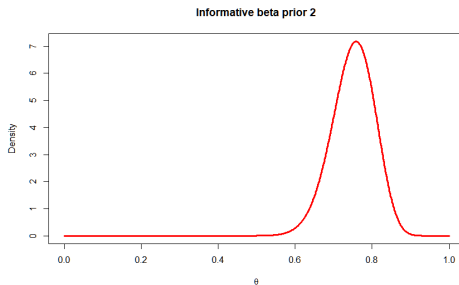
Prior model 2

An appropriate choice:

$$\theta \sim \text{Beta}(a = 45, b = 15)$$

Interpretation of prior hyperparameters:

- a = “prior number of successes” = 45
- b = “prior number of failures” = 15
- $a + b$ = “prior sample size” = 60



Computing posterior 2

We showed, using [Bayes' Theorem](#), that if $Y|\theta \sim \text{Bin}(n, \theta)$ and $\theta \sim \text{Beta}(a, b)$, then we get the following beta posterior:

$$\theta|y \sim \text{Beta}(a + y, b + n - y)$$

You will show in Homework 1 that the uniform prior we originally specified for θ is equivalent to a $\text{Beta}(a = 1, b = 1)$ distribution.

Thus, this result is a generalization of what we stated for posterior 1.

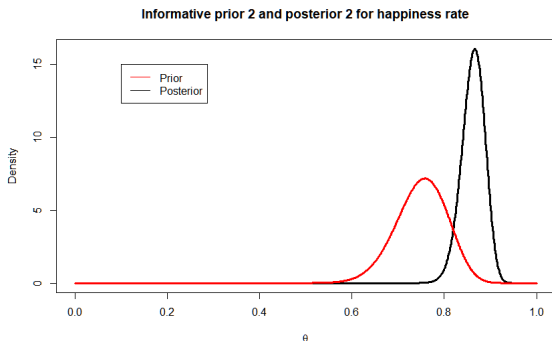
Visualizing posterior 2

Posterior 2 for female happiness example:

$$\theta|y = 118 \sim \text{Beta}(a + y = 45 + 118, b + n - y = 15 + 129 - 118)$$

$$\theta|y = 118 \sim \text{Beta}(163, 26)$$

We can visualize how the observed data updates our informative prior 2 beliefs into our posterior 2 beliefs about θ :



Numerical posterior summaries

It is useful to produce numerical summaries of the posterior distribution:

$$\theta \mid y \sim \text{Beta}(y + a, n - y + b).$$

First, we focus on measures of center.

Posterior mean: what is the expected proportion of generally happy females aged 65 or older, given the observed data?

$$\begin{aligned} E[\theta|y] &= \frac{a + y}{a + b + n} \\ &= \frac{a + b}{a + b + n} * \frac{a}{a + b} + \frac{n}{a + b + n} * \frac{y}{n} \\ &= \frac{a + b}{a + b + n} * \text{prior mean} + \frac{n}{a + b + n} * \text{sample mean} \end{aligned}$$

This is a weighted average of the prior mean and the sample mean

As the sample size $n \rightarrow \infty$, $E[\theta|y] \rightarrow y/n$ (the data is given more weight and “overrides” the prior info)

Numerical posterior summaries

Maximum a posteriori (MAP) estimate: what is the most likely proportion of generally happy females aged 65 or older, given the observed data?

$\hat{\theta}_{\text{MAP}}$ = value of θ which maximizes the posterior $p(\theta|y)$

For our posterior:

$$\hat{\theta}_{\text{MAP}} = \frac{a + y - 1}{a + b + n - 2}$$

since a $Beta(a, b)$ distribution with $a, b > 1$ is known to have mode $\frac{a-1}{a+b-2}$

Question: Why might we report the MAP estimate over the posterior mean?

- The MAP is the value of θ for which our posterior beliefs are highest

Numerical posterior summaries

We may also be interested in the amount of variation in the posterior of θ .

Posterior variance: how much uncertainty is there in the proportion of generally happy females aged 65 or older, given the observed data?

$$\text{Var}[\theta|y] = \frac{(a+y)(b+n-y)}{(a+b+n)^2(a+b+n+1)}$$

As the sample size of the observed data n increases, the posterior variance decreases

Numerical posterior summaries

A better way to quantify uncertainty in our posterior beliefs about θ is by computing a **range of values** for which it is highly likely θ falls within.

Definition: A $100(1 - \alpha)\%$ posterior credible interval for θ is determined by endpoints $l(Y)$ and $u(Y)$ which satisfy,

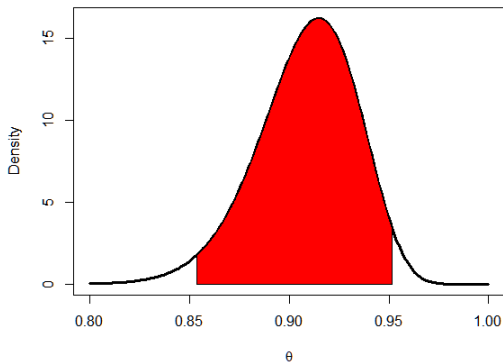
$$P(l(Y) < \theta < u(Y) | Y = y) = 1 - \alpha.$$

Note: This is not the same as a confidence interval!

Numerical posterior summaries

How do we select $l(Y)$ and $u(Y)$? One choice is to construct a **percentile-based interval**:

Zoomed-in posterior

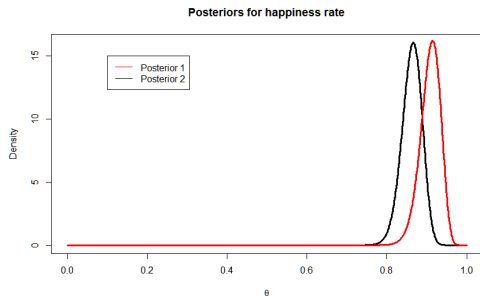


$$P(l(Y) < \theta < u(Y) | Y = y) = 1 - \alpha$$

Take $l(Y)$ to be percentile $100 * (\alpha/2)$ of $p(\theta|y)$ (in this case 0.854) and take $u(Y)$ to be percentile $100 * (1 - \alpha/2)$ of $p(\theta|y)$ (0.951)

Summarizing and comparing posteriors 1 and 2

Posterior	1: $\theta y \sim \text{Beta}(119, 12)$	2: $\theta y \sim \text{Beta}(163, 26)$
Posterior mean	0.908	0.862
Posterior variance	0.00063	0.000628
MAP	0.915	0.866
95% posterior CI	(0.854, 0.951)	(0.810, 0.908)



Posterior 2 shifted to left (informative prior centered around $\theta = 0.75$)

Concluding remarks

- Every Bayesian analysis proceeds in a similar fashion to this one.
- Neither posterior 1 nor 2 should be deemed **correct** or **incorrect** – depends on the amount of prior information available.
 - In this example, the posterior was quite different between the two prior choices.
 - Often beneficial to study how sensitive the posterior is to prior information – will discuss further
- **Generally, your prior distribution should not depend on the observed data!**
 - This is often referred to as “double-dipping,” and can lead to faulty conclusions due to over-reliance on the data.
 - You can base the prior on previous studies, scientific knowledge, or other intuition.

Bayesian Inference for the Binomial Model

Other Considerations

Topics

We will now start to address some finer details about the Bayesian machinery we have discussed so far. We will focus on the following:

- Shortcut for computing the posterior
- Conjugacy
- Types of credible intervals
- Sensitivity analysis

This is just a sampler of topics – we will discuss further modeling issues and considerations in the coming lectures.

Summary of Bayesian inference for binomial model

Recall the earlier example: In the 1998 General Social Survey, 118 ($= y$) of 129 ($= n$) females aged 65 or older claimed they were generally happy.

Goal: Estimate the true rate θ of generally happy females aged 65 or older.

- In Bayesian inference, our goal is to compute the **posterior distribution**, which represents our updated beliefs about θ **after** observing data y .
- $p(\theta)$ quantifies our prior beliefs about θ
- $p(y|\theta)$ likelihood of observed data
- $p(\theta|y)$ posterior distribution (updated beliefs about θ)

Summary of Bayesian inference for binomial model

Recall: Let Y be a random variable counting the number of “successes” within a sample of size n , and θ represents the true rate of “success” in the population.

- Sampling model:

$$Y|\theta \sim \text{Bin}(n, \theta)$$

$$p(y|\theta) = \binom{n}{y} \theta^y (1 - \theta)^{n-y} \text{ for } y = 0, 1, \dots, n$$

- Prior model:

$$\theta \sim \text{Beta}(a, b)$$

$$p(\theta) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1} (1-\theta)^{b-1} \text{ for } 0 \leq \theta \leq 1$$

(“noninformative” prior: $\text{Unif}[0, 1]$, which is equivalent to $\text{Beta}(1, 1)$)

Summary of Bayesian inference for binomial model

We compute the **posterior distribution** via **Bayes' Theorem**:

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)},$$

where $p(y) = \int_{-\infty}^{\infty} p(y|\theta)p(\theta) d\theta$ is the *normalizing constant*.

Previously in class, we showed that the posterior is:

$$\theta|y \sim \text{Beta}(a + y, b + n - y)$$

$$p(\theta|y) = \frac{\Gamma(a + b + n)}{\Gamma(a + y)\Gamma(b + n - y)} \theta^{a+y-1} (1 - \theta)^{b+n-y} \text{ for } 0 \leq \theta \leq 1$$

A shortcut for computing the posterior

A key part of deriving the posterior is computing the **normalizing constant**:

$$\begin{aligned} p(y) &= \int_{-\infty}^{\infty} p(\theta)p(y|\theta)d\theta \\ &= \binom{n}{y} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \int_0^1 \theta^{(a+y)-1} (1-\theta)^{(b+n-y)-1} d\theta \end{aligned}$$

- This is difficult to compute directly
- In class, we used the **"kernel trick"** to evaluate the integral, by recognizing that the inside of the integral looked like a $\text{Beta}(a+y, b+n-y)$ density

Question: Do we actually need to compute this constant?

A shortcut for computing the posterior

Consider: Suppose we are interested in estimating θ by either θ_1 or θ_2 . Given data y , which value is a better estimate?

We would prefer θ_1 if $p(\theta_1|y) > p(\theta_2|y)$, or

$$\frac{p(\theta_1|y)}{p(\theta_2|y)} > 1$$

But notice that

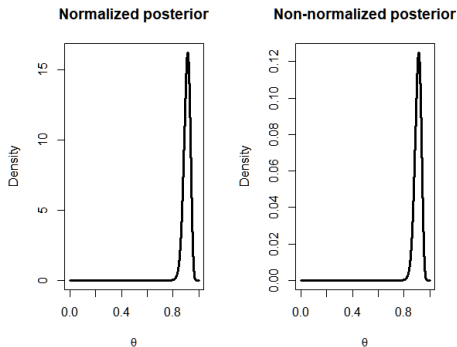
$$\frac{p(\theta_1|y)}{p(\theta_2|y)} = \frac{p(y|\theta_1)p(\theta_1)/p(y)}{p(y|\theta_2)p(\theta_2)/p(y)} = \frac{p(y|\theta_1)p(\theta_1)}{p(y|\theta_2)p(\theta_2)}$$

does not depend on the normalizing constant $p(y)$

This means the posterior's **shape** only depends on the numerator

A shortcut for computing the posterior

The posterior's **shape** only depends on the numerator:



Left: $p(\theta|y) = \frac{p(\theta)p(y|\theta)}{p(y)}$ **Right:** $p(\theta)p(y|\theta)$

Notice that they have the same shape. If $\theta_2 = 0.9$ and $\theta_1 = 0.3$, we would prefer θ_2 over θ_1 no matter which plot we look at.

A shortcut for computing the posterior

In light of the previous slide, a simpler way to compute the posterior in this case is to work with **proportionality statements** (everything that does not depend on θ is treated as a constant):

$$\begin{aligned} p(\theta|y) &= \frac{p(\theta)p(y|\theta)}{p(y)} \propto p(\theta)p(y|\theta) \\ &\propto \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1} (1-\theta)^{b-1} \binom{n}{y} \theta^y (1-\theta)^{n-y} \\ &\propto \theta^{(a+y)-1} (1-\theta)^{(b+n-y)-1} \end{aligned}$$

We recognize the last expression as being proportional to the probability density function of a $\text{Beta}(a+y, b+n-y)$ distribution.

Thus, we can conclude that the posterior distribution is in fact $\text{Beta}(a+y, b+n-y)$ (Why?)

A shortcut for computing the posterior

If $p(\theta|y) \propto \theta^{(a+y)-1}(1-\theta)^{(b+n-y)-1}$, then it is proportional to the probability density function

$$f(\theta) = \frac{\Gamma(a+b+n)}{\Gamma(a+y)\Gamma(b+n-y)} \theta^{(a+y)-1} (1-\theta)^{(b+n-y)-1}$$

of a $\text{Beta}(a+y, b+n-y)$ distribution.

This implies that there exists a constant c such that $p(\theta|y) = c * f(\theta)$.

Since $p(\theta|y)$ and $f(\theta)$ are probability density functions, each must integrate to 1. Therefore:

$$1 = \int_{-\infty}^{\infty} p(\theta|y) d\theta = \int_{-\infty}^{\infty} c * f(\theta) d\theta = c \int_{-\infty}^{\infty} f(\theta) d\theta = c * 1 = c.$$

So $c = 1$ and $p(\theta|y) = f(\theta)$.

Conjugacy

Observation: For the binomial model with beta prior, we were able to **directly** recognize the posterior distribution. Does this always happen?

Suppose we specified the following normally distributed prior instead:

$$p(\theta) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(\theta-\mu)^2/2\sigma^2}, \text{ for } -\infty < \theta < \infty$$

(We can choose μ and a small σ to concentrate the mass in $[0, 1]$)

Then

$$\begin{aligned} p(\theta|y) &\propto p(\theta)p(y|\theta) \\ &\propto \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(\theta-\mu)^2/2\sigma^2} \binom{n}{y} \theta^y (1-\theta)^{n-y} \\ &\propto e^{-(\theta-\mu)^2/2\sigma^2} \theta^y (1-\theta)^{n-y} \end{aligned}$$

Unfortunately, this is not proportional to the probability density function of a distribution that we recognize.

Conjugacy

The prior $\theta \sim \text{Beta}(a, b)$ was conveniently chosen because we obtain a closed-form posterior distribution.

Definition: Let $\mathcal{F}$ be the class of sampling models $p(y|\theta)$ and $\mathcal{P}$ be the class of prior distributions $p(\theta)$. We say the prior class $\mathcal{P}$ is **conjugate** for sampling model class $\mathcal{F}$ if:

$$p(\theta) \in \mathcal{P} \text{ and } p(y|\theta) \in \mathcal{F} \text{ implies } p(\theta|y) \in \mathcal{P}.$$

In other words, prior $p(\theta)$ and posterior $p(\theta|y)$ come from the same family of distributions

Conjugacy

Example: Binomial model

- **prior:** $\theta \sim \text{Beta}(a, b)$
- **posterior:** $\theta|y \sim \text{Beta}(a + y, b + n - y)$

A beta prior is conjugate for the binomial model because the posterior distribution is also a beta distribution

Conjugacy

Pros to conjugate prior specification:

- Allows *direct* computation of posterior (i.e., a closed-form, recognizable posterior distribution)
- Easy-to-understand and interpret
- Convenient starting choice

Cons to conjugate prior specification:

- Not always available for every model (particularly if θ consists of a large number of parameters, i.e., high-dimensional)
- Restrictive in terms of modeling choices – might not always be the choice which best reflects prior beliefs about θ

Types of credible intervals

Recall: We can quantify uncertainty in our posterior estimation of θ by computing a **posterior credible interval** for which it is likely θ falls within.

A $100(1 - \alpha)\%$ posterior credible interval is $[l(Y), u(Y)]$, where

$$P(l(Y) < \theta < u(Y) | Y = y) = 1 - \alpha$$

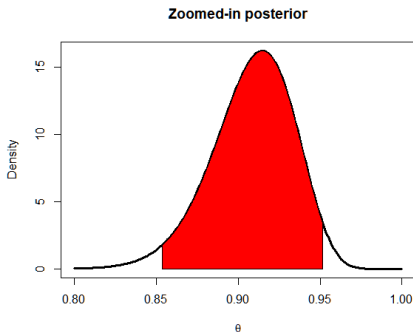
Question: How do we select $l(Y)$ and $u(Y)$?

Two different ways to construct intervals by determining these endpoints:

- **Percentile-based interval** – discussed last time
- **Highest posterior density (HPD) region** – will discuss this time

Types of credible intervals

(1) Percentile-based interval:

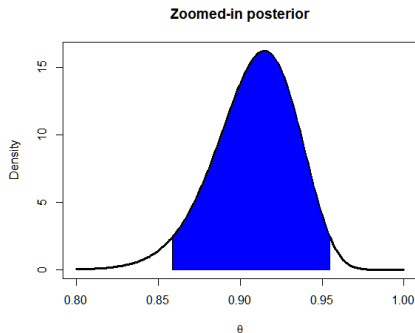


Pros: Easy to compute, most commonly used.

Con: Some values of θ outside of a percentile-based credible interval have higher posterior density than values within!

Types of credible intervals

(2) Highest posterior density (HPD) region: resolves the con of percentile-based intervals



Pro: Values within the HPD region have higher posterior density than values outside.

Con: Can be difficult to compute.

HPD Intervals

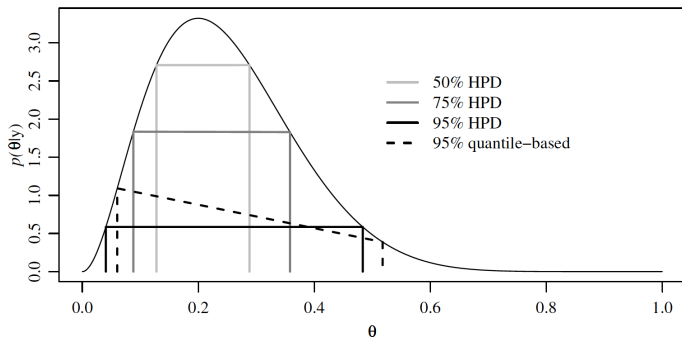
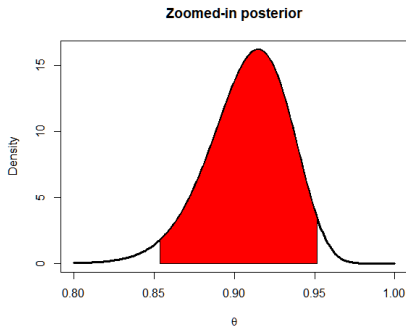


Fig. 3.6. Highest posterior density regions of varying probability content. The dashed line is the 95% quantile-based interval.

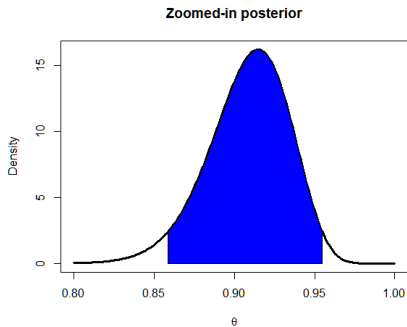
As shown in Hoff's text: the basic idea is to gradually move a horizontal line down across the density, including in the HPD region all values of θ having a density above the horizontal line. Stop moving the line down when the posterior probability of the θ -values in the region reaches $1 - \alpha$.

Types of credible intervals

Female happiness example: assuming the prior $\theta \sim \text{Unif}(0, 1) = \text{Beta}(1, 1)$



Left: 95% percentile-based CI: (0.854, 0.951)



Right: 95% HPD region: (0.858, 0.955)

Sensitivity analysis

Re-visiting prior specification: Suppose we believed from past studies that the true rate of generally happy females aged 65 or older was close to 75%. Consider the conjugate prior for θ , with the following hyperparameter choices:

Prior	"Prior" sample size	a	b
(1)	2	1	1
(2)	4	$4(0.75) = 3$	1
(3)	60	45	15
(4)	120	90	30

$$\theta \sim \text{Beta}(a, b) \text{ and } E(\theta) = \frac{a}{a+b}$$

We think of a as the prior number of "successes," b as the prior number of "failures," and $a + b$ as the prior "sample size"

Prior (1) is "noninformative" (equivalent to $\text{Unif}[0, 1]$) while the others are centered at 0.75

Sensitivity analysis

The posterior distribution under prior choices (1)–(4):

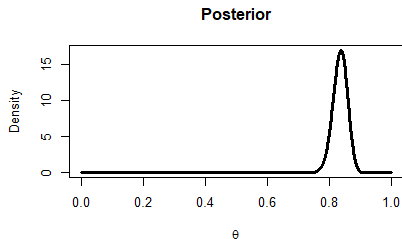
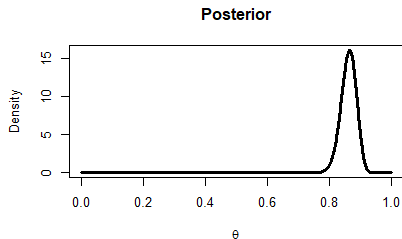
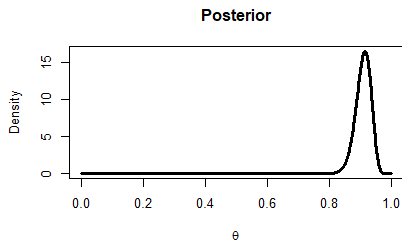
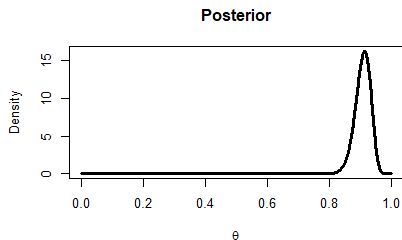
- (1): $\theta|y = 118 \sim \text{Beta}(119, 12)$
- (2) $\theta|y = 118 \sim \text{Beta}(121, 12)$
- (3) $\theta|y = 118 \sim \text{Beta}(163, 26)$
- (4) $\theta|y = 118 \sim \text{Beta}(208, 41)$

A study of how sensitive the posterior is to our prior assumptions is called a **sensitivity analysis**.

- **Remarkably important** for any Bayesian analysis!

Sensitivity analysis

Plot of posteriors (as we go from (1) to (4) we place more weight on prior beliefs)



Sensitivity analysis

95% HPD credible intervals for θ :

Prior	Credible interval
(1)	(0.858, 0.955)
(2)	(0.860, 0.955)
(3)	(0.813, 0.910)
(4)	(0.789, 0.880)

Take-away messages:

- If one does not have any prior information, using a weakly informative prior to “let the data drive inference” is a good choice.
- If one does have prior information, use it carefully and be able to justify! Especially important if the posterior is especially sensitive to the choice of prior (i.e., if you do not have much observed data).